

EXPERIMENT NO: 1

AIM: To determine natural moisture content of soil using Oven Drying Method.

APPARATUS:

1. Containers (non corrodible, air-tight)
2. Balance (accuracy 0.04 percent of the weight of the soil taken for test).
3. Oven (interior of non-corroding material, thermostatically controlled).
4. Desiccators.
5. Tongs (one pair).

THEORY: A soil is an aggregate of soil particles having a porous structure. The pores may have water and/or air. The pores are also known as voids. If voids are fully filled with water , the soil is called saturated soil and if voids have only air, the soil is called dry. Moisture content is defined as the ratio of the mass/weight of water to the mass/weight of soil solids.

PROCEDURE:

1. Clean, dry and weigh the container with lid.
2. Take the required quantity of the soil specimen in the container and weigh with lid.
3. Maintain the temperature of the oven between 105° C to 110° C for normal soils and 600°C to 800 °C for soils having loosely bound hydration water or/and Organic matter.
4. Dry the sample in the oven till its mass becomes constant. In normal conditions the sample is kept in the oven for not more than 24 hours.
5. After drying remove the container from the oven, replace the lid and cool in the desiccators.
6. Weigh the dry soil in the container with lid

Precautions

1. The soil specimen should be loosely placed in the container.
2. Over heating should be avoided.
3. Dry soil sample should not be left in open before weighing.

Observations and Calculations

The moisture content is calculated as follows:

$$W = W_w/W_s = (W_2 - W_3) / (W_3 - W_1) \times 100$$

Where , W_1 = mass of container with lid

W_2 = mass of container with lid + wet soil

W_3 = mass of container with lid + dry soil

Sl no	Particulars	1	2
1	Container No.		
2	Mass of empty container with lid W ₁ (gm)		
3	Mass of container with lid + wet soil, W ₂ (gm)		
4	Mass of container with lid + dry soil, W ₃ (gm)		
5	05 Mass of water, W _w = W ₂ - W ₃ (gm)		
6	Mass of dry soil, W _s = W ₃ - W ₁ (gm)		
7	Moisture content, W = W ₂ - W ₃ / W ₃ -W ₁ x 100%		

Result -

Average Moisture Content, w = (%)

EXPERIMENT NO: 2

AIM: To determine the specific gravity of the soil particles passing 4.75 mm I.S. sieve using pycnometer.

INSTRUMENTS REQUIRE -

1. Pycnometer
2. 4.75 mm sieve
3. Vacuum pump (or hot water bath)
4. Balance (accuracy 0.1 gm)
5. Drying oven
6. Desiccator
7. Glass rod
8. Distilled and deaired water
8. Thermometer (0 to 500 C)

Theory - Specific gravity is the ratio of the mass/weight in air of a given volume of dry soil solids to the mass/weight of equal volume of distilled water at 4° C.

Where, empty pycnometer mass = W_1

pycnometer + soil grains mass = W_2

pycnometer + soil grains + water mass = W_3

pycnometer + water mass = W_4

The value of specific gravity depends upon temperature, hence its value is reported at standard temperature of 27°C

PROCEDURE:

1. Dry the pycnometer thoroughly and weigh with its cap tightly screwed on.
2. Mark the cap and pycnometer with a vertical line parallel to the axis of pycnometer so that each time the cap is screwed the same amount.
3. Unscrew the cap and put in about 200 gm of oven dried soil passing 4.75 mm I.S. sieve and weigh again.
4. Add sufficient deaired water to cover the soil about half full and screw on the cap.
5. Shake well and connect to the vacuum pump to remove entrapped air.
6. Allow the air to be evacuated for at least 20 minutes for grained soils and 10 minutes for sandy soils. Shake the pycnometer occasionally to assist in the air evaluation.
7. After the entrapped air has been largely removed, disconnect the pump and fill the pycnometer with water about three fourth full.

8. Reapply the vacuum for at least 5 minutes. Evacuation should be continued until very few bubbles appear on the top of the water.
9. After the air has been eliminated, full the pycnometer with water completely up to the mark.
10. Thoroughly dry the pycnometer from the outside and weigh it.
11. Record the temperature of the content in degree centigrade.
12. Clean the pycnometer by washing water thoroughly.
13. Fill the pycnometer with water up to its top and screw on the cap.
14. Weigh the pycnometer after drying it on the outside thoroughly.
15. Repeat the test twice more.

Precautions

1. The soil grains whose specific gravity is to be determined should be completely dry.
2. Dried soil taken for testing should have the soil grains of its original size, so if on drying soil humps are formed, they should be broken to its original size.
3. Hold the rubber tubing tightly with the pycnometer so that there is no leakage when Vacuum pump works.
4. Inaccuracies in weighing and failure to completely eliminate the entrapped air are the main source of error. Both should be avoided by careful working.
5. Cap should be screwed up to the same mark during the test.
6. Cap should be properly screwed with washer to avoid any leakage.

OBSERVATION TABLE-

Determine the specific gravity of soil grains using equation

$$G = (W_2 - W_1) / (W_2 - W_1) - (W_3 - W_4)$$

Result

Specific Gravity of Soil =

EXPERIMENT NO: 3

AIM: To determine the field density of soil by Core cutter method

APPARATUS REQUIRED :

- a) Cylindrical core cutter (height = 12.74 cm, dia 10 cm)
- b) Steel rammer
- c) Steel dolly (2.5 cm high and 10 cm internal diameter)
- d) Balance (accuracy 1gm)
- e) Steel rule
- f) Spade or pickaxe
- g) Straight edge
- h) Knife
- i) Water content crucibles
- j) Desiccator
- k) Oven
- l) Tongs

Theory:- Density is defined as the mass per unit volume of soil

$$\gamma = W/V$$

Where γ = mass density of soil

w = total mass of soil

v = total volume of soil

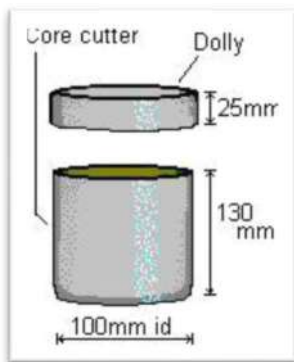
Density is used in calculating the stress in the soil due to its overburden pressure. It is needed in estimating the bearing capacity of soil foundation system, settlement of footings, earth pressures behind the retaining walls, dams, embankments. Stability of natural slopes, dams, embankments and cuts is checked with the help of density of these soils. It is the density of controls the field of soils. Permeability of soils depends upon its density. Relative density of cohesion less soils is determined by knowing the dry density of that soil in natural, loosest and densest states. Void ratio, porosity and degree of saturation need the help of density of soil. The density can be expressed in g/cm³, or t/m³ or kg/m³ or lb/t³. For calculating the submerged density the density of water is taken as 1 g/cm³ = 1 t/m³. Dry density of the soil is calculated by using equation –

$$\gamma_d = \gamma_t / (1 + w)$$

Where, γ_d = dry density of soil

γ_t = wet density of soil

w = moisture content of soil



Precautions

1. Steel dolly should be placed on the top of the cutter before ramming it down.
2. Core cutter should not be used in gravels and boulders.
3. Before lifting the cutter, soil should be removed round the cutter, to minimize the disturbances.
4. While lifting the cutter, no soil should drop down,
5. During pressing and lifting the cutter care should be taken that some soil is projected at both the ends of the cutter.
6. Values should be reported to second place of decimal .

Observations and calculation

Internal Diameter of Cutter (cm) =

Height of Cutter (cm) =

Cross-sectional Area of cutter (cm²) =

Volume of Cutter (cm³) =

Specific Gravity of soil =

Sl no	particulars	Test 1	Test 2	Test 3
1	Wt of empty cutter(w ₁) gms			
2	Wt of cutter+wet soil(w ₂) gms			
3	Volume of core cutter, V _c , gm/cc			
4	Wt of empty container(w ₃) gms			
5	Wt of container+wet soil(w ₄)			
6	Wt of container+dry soil(w ₅)			
7	Water content(w)			
8	Field moist density, $\gamma_t = (w_2 - w_1)/V_c$ (gm/cc)			
9	Dry density $\gamma_d = \gamma_t / (1 + w)$, gm/cc			
10	Avg density (γ_{avg}) gm/cc			

Result- the average in-situ field density= _____ gm/cc.

EXPERIMENT NO: 4

AIM: To determine the field density of soil by sand replacement method.

APPARATUS REQUIRED :

1. Sand pouring cylinder
2. Trowel or bent spoon
3. Cylindrical calibrating container
4. Metal tray with hole (30 cm square with 10 cm hole in the centre)
5. Sand (clean oven dried, passing 600 micron sieve)
6. Balance (accuracy 1gm)
7. Moisture content crucibles
8. Oven
9. Desiccator
10. Tongs
11. Glass plate (about 45 cm square)
12. Metal tray (about 30 cm square)
13. Scraper tool
14. Measuring jar

Procedure -

Calibration of Apparatus

1. Measure the internal volume of the calibrating container from the volume of the water required to fill the container.
2. Fill the pouring cylinder with sand within about 1.0 cm of the top and weight it.
3. Place the pouring cylinder concentrically on the top the calibrating container.
4. Open the shutter to allow the sand to run out and fill calibrating cylinder.
5. When there is no further movement of sand in the pouring cylinder closed the shutter.
6. Remove the pouring cylinder and weight it to the nearest gram.
7. Place the pouring cylinder on a plane surface such as the glass plate
8. Open the shutter and allow the sand to run out. When there is no movement of sand in the cylinder, close the shutter.
9. Weigh the pouring cylinder with remaining sand.

Measurement of Soil Density

1. Clean and level the ground where the field density is required.
2. Fill the pouring cylinder with dry sand within about 1.0 cm of the top and weigh it.
3. Place the metal tray with the central hole over the portion of soil to be tested.
4. Excavate the soil approximately 10 cm dia and 15 cm deep with bent spoon. The hole in the tray will guide the diameter of the hole to be made in the soil.
5. Collect the excavated soil in the metal tray, weight it to the nearest gram.
6. Determine moisture content of the excavated soil.

7. Place the pouring cylinder over the hole so that base of the cylinder over Concentrically.
8. Open the shutter and allow the sand to run out into the hole. When there is no Movement of sand, the shutter is closed.
9. Remove the cylinder and weight it.

Observations and Calculation

1. Enter all the readings in table 2, 3 and 4

2. Bulk density of sand is calculated as shown in table 2. This density is used in determining the volume of the hole made in the soil.

3. Table 4 show the calculations of wet density, dry density, void ratio and degree of saturation of the soil.

4. Above Equations are used to calculate the dry density, void and degree of saturation

TABLE-2: Calibration of apparatus

Sl no	Description	1	2
1	Container No.		
2	Mass of container with lid. W ₁ (gm)		
3	Mass of container with lid + wet soil, W ₂ (gm)		
4	Mass of container with lid + dry soil, W ₂ (gm)		
5	Mass of water, W _w = W ₂ - W ₃ (gm)		
6	Mass of dry soil, W _s = W ₃ - W ₁ (gm)		
7	Moisture content, W = $[W_2 - W_3 / W_3 - W_1] \times 100$, (%)		

TABLE-3: Water content determination

Sl no	description	1	2
1	Volume of Calibrating Container, V (gm)		
2	Mass of Pouring Cylinder+ Sand, W' ₁ (gm) (Before Pouring In the Calibration Cylinder)		
3	Mass of Pouring Cylinder +Sand W' ₂ (gm) (After Pouring In the Calibration Cylinder)		
4	Mass of Pouring Cylinder +Sand W' ₃ (gm) (After Making the sand cone on flat surface)		
5	Mass of sand for filling the calibrating cylinder and cone (gm) W' ₄ = (W' ₁ -W' ₂)		
6	Mass of sand for making the cone only (gm) W' ₅ = (W' ₂ -W' ₃)		
7	Mass of sand in the calibrating cylinder only (gm) W' ₆ = (W' ₄ -W' ₅)		
8	Bulk Density of Sand , $\gamma_d = W'_6 / V$ (gm/cc)		

TABLE-4: Sand replacement method data

Sl no	description	1	2
1	Mass of Pouring Cylinder+ Sand, W'1(gm) (Before Pouring In the Calibration Cylinder)		
2	Mass of Pouring Cylinder +Sand W'2 (gm) (After Pouring In the Calibration Cylinder)		
3	Mass of Pouring Cylinder +Sand W'3 (gm)		
4	Mass of sand used in the hole and cone(gm) $W4 = (W1 - W2)$		
5	Mass of sand in the Cone only (gm) $W5 = (W2 - W3)$		
6	Mass of sand in the hole only (gm) $W6 = (W4 - W5)$		
7	Volume of Sand, $V = W6/\gamma_d$ (from Table 2)		
8	Mass of Tray + Excavated Soil, W7 (gm)		
9	Mass of Tray Only, W8 (gm)		
10	Mass of Excavated Soil, (W= W7-W8) (gm)		
RESULT-			
1	Wet Density (γ_t) = $(10 / 7)$ (gm/cc)		
2	Dry Density (γ_d) = $\gamma_t / (1+w)$ (gm/cc)		
3	Void Ratio, $e = [(G_s \gamma_w) / \gamma_d] - 1$		
4	Degree of Saturation, $S = (w.G_s / e) \times 100$ (%)		

Result:

In-situ density of soil by

1. Core cutter = ----- gm/cc

2. Sand replacement = ----- gm/cc

EXPERIMENT NO :- 5

Aim – To determine grain size distribution by sieve analysis .

Apparatus –

1. 1st set sieves of sizes 300 mm, 80 mm, 40 mm, 20 mm, 10 mm, and 4.75 mm.
2. 2nd set of sieves of sizes 2 mm, 850 μ , 425 μ , 150 μ , and 75 μ .
3. Mechanical shaker (optional)
4. Brush
5. Balances (one of accuracy=1.0 gm. Other of accuracy=0.1 gm)
6. Oven
7. Desiccator
8. Drying crucibles
9. Tray/bucket

Theory: Soils having particles larger than 0.075mm size are termed as coarse grained soils. In these soils more than 50% of the total material by mass is larger than 75 micron. Coarse grained soil may have boulder, cobble, gravel and sand.

Coarse : 4.75mm to 2mm

Medium : 2.0mm to 425 micron

Fine : 425 micron to 75 micron

Name of the soil is given depending upon the maximum percentage of the above components. Soils having less than 5% particles of size smaller than 0.075 mm are designed by the symbols:

GW: Well graded gravel

GP-Poorly graded gravel

SW: Well graded sand

SP-Poorly graded sand

Soils having greater than 12% of particles of size smaller than 0.075 mm are designed by the following symbols:

GM or GC: Silty gravel or clayey sand

SM or SC: Silty sand or clayey sand

Coefficient of curvature may be estimated as:-

$$Cc = (D_{30})^2 / (D_{10} \times D_{60})$$

Where,

D_{60} =diameter at 60% finer

D_{30} =diameter at 30% finer

D_{10} =diameter at 10% finer

It should lie between 1 and 2 for well graded gravels and sands

Uniformity coefficient

$$Cu = D_{60} / D_{10}$$

Procedure-

1. Take suitable quantity of oven-dry soil depending upon the maximum size of material present in.
2. Sieve the soil first through I.S. sieves of first set i.e. 80 mm, 40 mm, 10mm and 4.75 mm manually or using a mechanical shaker for 5-10 minutes.

3. Weigh the material retained on each sieve to 1.0 gm.
4. Sieve the soil through 2nd set of sieves i.e., 2mm, 600 μ , 425 μ , 150 μ and 75 μ using a mechanical shaker for 10 minutes.
5. Weigh to 0.1 gm each sieve and pan with soil retained on them. The sum of the retained soil mass is checked against the original mass of soil taken.

Precaution-

1. While drying the temperature of the oven should not be more than 105°C because higher temperature may cause some permanent change in the -75 μ materials.
2. During shaking, sample soil should not be allowed to come out.
3. In wet analysis, all cohesive soil adhering to large size particles should be removed by water.
4. For plotting, per cent finer should be determined with respect to the total soil taken for initial analysis.

Calculation-

Sl no	Sieve no	Mass of soil retained(gm)	Cum mass of soil retained(gm)	Cum %of soil retained	% finer or passing
1	4.75mm				
2	2.36mm				
3	1.18mm				
4	600micron				
5	300micron				
6	150micron				
7	75micron				
8	pan				

Draw Graph sheet – Grain Size Analysis

Result

1. Coefficient of Curvature, C_c =
2. Uniformity Coefficient, C_u =

EXPERIMENT NO :- 6

Aim – To determine liquid limit, plastic limit, shrinkage limit of soil sample.

Determination of liquid limit-

Apparatus:-

1. Cassagrande's liquid limit device
2. A.S.T.M. and B.S. grooving tool (Cassa grande's type)
3. Glass plate
4. 425 micron I.S. sieve
5. 3mm diameter rod.
6. Spatula
7. Balance (0.01 gm sensitivity),
8. Water content tins or crucibles
9. Drying oven
10. Distilled water
11. Measuring cylinder
12. Desiccator

Theory:- Liquid limit is the water content at which soil passes from zero strength to an Infinitesimal strength, hence the true value of liquid limit can not be determined. For determination purpose liquid limit is that water content at which a part of soil, cut by a groove of standard dimensions, will flow together for a distance of 1.25cm under an impact of 25 blows in a standard liquid limit apparatus.

The soil at the water content liquid limit apparatus . **Procedure for liquid limit:-**

- 1) Adjust the cup of the liquid limit apparatus with the help of grooving tool gauge & the adjustment plate to give a drop of exactly 1 cm on the point of contact on base.
- 2) Take about 120gm of an air dried sample passing 425 μ sieve .
- 3) Mix it thoroughly with some distilled water to form a uniform paste.
- 4) Place a portion of the paste in the cup of the liquid limit device, smooth the surface with spatula to a maximum depth of 1cm. Draw the grooving tool through the sample along the symmetrical axis of the cup, holding the tool perpendicular to the cup.
- 5) Turn the handle at a rate of 2 revolutions per second and count blows until two parts of the soil sample come into contact at the bottom of the soil sample come into contact at the bottom of the groove along a distance of 10mm.
- 6) Transfer about 15gm of the soil forming the edges of the groove that flowed together to a water content tin and determine the water content by oven drying.
- 7) Transfer the remaining soil in the cup to the main soil sample in the basin and mix thoroughly after adding a small amount of water

- 8) Repeat steps 4, 5 and 6. Obtain at least four sets of readings in the range of 10 to 40 blows.

Observation and Calculations

(a) Liquid Limit (L.L. or WL.L)

1. Use table 1 for recording the number of blows and calculating the moisture contents.
2. Use semi log graph paper, take number of blows on semi log scale (x-axis) and water contents on ordinary scale (y-axis). Plot all the points and draw a straight line (flow curve) passing through these points.
3. Read the water content at 25 blows which is the value of liquid limit.

Sl no	description	1	2	3
1	No. of blows			
2	Container No.			
3	Mass of container + wet soil (gm)			
4	Mass of container + dry soil ,(gm)			
5	Mass of water (3)-(4) (gm)			
6	Mass of container, (gm)			
7	Mass of dry soil (4)-(6)			
8	Moisture content (5)/ (7) x 100, (%)			

Liquid limit: ...

Procedure for plastic limit:-

- 1) Take about 30gm of air dried sample passing 425 micron sieve.
- 2) Mix thoroughly with distilled water on the glass plate until it is plastic enough to be shaped into a small ball.
- 3) Take about 10gm of the plastic soil mass and roll it between the hand and the glass plate to from the soil mass into a thread. If diameter of thread becomes less than 3mm without cracks, it shows that water added in the soil is more than its plastic limit; hence the soil is kneaded further and rolled into thread again.
- 4) Repeat this rolling and remolding process until the thread starts just crumbling at a diameter of 3mm.
- 5) If crumbling starts before 3mm diameter thread in step 3, it show the water added in step 2 is less than the plastic limit of the soil, hence some more water should be
- 6) added and mixed to a uniform mass and rolled again, until the thread starts just crumbling at a diameter of 3mm.
- 7) Collect the pieces of crumbled soil thread at 3mm diameter in an air tight container and determine moisture content.
- 8) Repeat this procedure twice more with fresh samples of 10gm each.

Observation table-

Sl no	description	1	2	3
1	No. of blows			
2	Container No.			
3	Mass of container + wet soil (gm)			
4	Mass of container + dry soil ,(gm)			

5	Mass of water (2)-(3) (gm)			
6	Mass of container, (gm)			
7	Mass of dry soil (3)-(5) gm			

Average plastic limit:

Result-

For **Classification of soil-**

1. Calculating plasticity index (Ip)

$$I_p = W.L.L - W.P.L.$$

Calculate the plasticity the plasticity index of 'A' line

$$(P.I.)_A = 0.73 (W.L.L - 20)$$

Where W.L.L is in percentage

If $P.I. > (P.I.)_A$ the soil is clay

If $P.I. < (P.I.)_A$ the soil is silt

L.L. = 0-35 low Compressibility

35-50 medium Compressibility

> 50 High Compressibility

Determination of shrinkage limit of soil-

Apparatus-

1. Three circular shrinkage dish (porcelain / stainless steel/brass with flat bottom about 4.5 cm in dia and 1.5 cm high)
2. Three porcelain evaporated dish (two about 12cm large) and one 6cm (small in diameter)
3. One glass plate with three prongs.
4. Plain glass plate (7.5 cm X 7.5 cm).
5. One glass or stainless steel cup (about 5.0 cm in dia and 2.5 cm high with level and smooth ground top rim).
6. Mercury
7. 425 micron sieve .
8. Spatula
9. Straight edge
10. Oven
11. Desiccator
12. Balance (sensitivity 0.01 gm)

Theory:

Shrinkage limit is the maximum water content at which a reduction in water content does not cause an appreciable decrease in volume of the soil mass. At shrinkage limit, on further reduction in water, air starts to enter into the voids of the soil and keeps the volume of voids constant.

Procedure

1. Mix about 30gm of soil passing 425 micron sieve with distilled water. The water added should be sufficient to make the soil past enough to be readily worked into the shrinkage dish without inclusion of air bubbles.
2. Coat the inside of two shrinkage dish with a thin layer of Vaseline. Place the soil sample in the dish, by giving gentle taps. Strike off the top surface with a straight edge.
3. Weight the shrinkage dish immediately full of wet soil. Dry the dish first in air and then in an oven.
4. Weight the shrinkage dish with dry soil pat.
5. Clean dry the shrinkage dish and determine its empty mass.
6. Also weigh an empty porcelain dish (small size) which will be used for weighing dish.
7. Keep the shrinkage dish in a large porcelain dish, fill it to overflowing with mercury and remove the excess by pressing the plate firmly over the top of the dish. Transfer the contents of the shrinkage dish to the mercury weighing dish and weight.
8. Place the glass cup in a large dish. Fill it to over flowing with mercury; remove the excess by pressing the glass plate with three prongs firmly over the top of the cup.
9. Wipe the outside of the glass cup to remove any adhering mercury, and then place it in another large dish. Place the dry soils pat on the surface of the mercury and submerge it under the mercury by pressing with glass plate with prongs.
10. Transfer the mercury displaced by the dry soil pat to the mercury weighing dish and weigh.
11. Repeat the test at least three times for each soil sample.

Precautions

1. The water content of the soil taken in shrinkage dish should be above liquid limit but within 10% from liquid limit.
2. To prevent the cake from adhering to the shrinkage dish and consequent cracking of the dry soil pat, the inside of the shrinkage dish should be greased with Vaseline.
3. During filling the shrinkage dish with soil paste, sufficient tapping should be done to remove the entrapped air.
4. The dry soil pat should be weighed soon after it has been removed from the air.
5. Test should be repeated at least three times for each soil sample and the average of the results thus obtained is reported.
6. No air should be entrapped under the dry soil pat when pressing by the glass with prongs is being carried out.

description	1	2	3
Shrinkage dish no. (gm)			
Mass of dish+ wet soil pat (gm)			
Mass of dish+ dry soil pat (gm)			

Mass of water,(2)-(3) (gm)			
Mass of shrinkage dish empty (gm)			
Mass of dry soil pat (Ws)=(3)-(5) (gm)			
Initial water content (%)			
Mass of weighing dish +mercury (filling shrinkage dish)			
Mass of weighing dish empty (gm)			
Mass of mercury (8)-(9) (gm)			
Vol. wet soil pat (cc)			
Mass of weighing dish+ displaced mercury (by dry pat)			
Mass of mercury displaced (12) – (9)			
Volume of dry soil packed			

$$W_{SL} = (W_1(W_1 - W_s) - (V_1 - V_2) \gamma_w) W_s \times 100$$

Where,

W S.L. = shrinkage limit in % soil pat.

W1 = initial water content of wet soil pat. V1 = volume of wet soil pat in cc.

V2 = volume of dry soil in cc.

Ws = mass of oven-dry soil pat in gm. γ_w = mass density of water in g/cc.

EXPERIMENT NO :- 7

Aim – To perform Permeability test using Constant-head test method.

Apparatus:-

1. Permeameter mould (internal dia = 100 m.m. effective height = 127.3 m.m. Capacity 100cc)
2. Accessories of the permeameter (cover, base, detachable collar, porous stones, dummy plate) (common)
3. Round filter paper 100 mm. dia (common)
4. A static or dynamic compaction device (if remolded samples are used)
5. Constant head reservoir (common)
6. Graduated glass stand pipe (internal dia 5to 20 mm, preferably 10mm) (Variable head)
7. Support frame and clamps (variable head)
8. Funnel (variable head)
9. Measuring flask (Constant head)
10. Metre Scale
11. Balance
12. Stop watch
13. Thermometer
14. Deaired water
15. Drying oven
16. IS: Sieve 4.75 mm
17. Grease
18. Straight edge
19. Drying crucibles
20. Desiccator

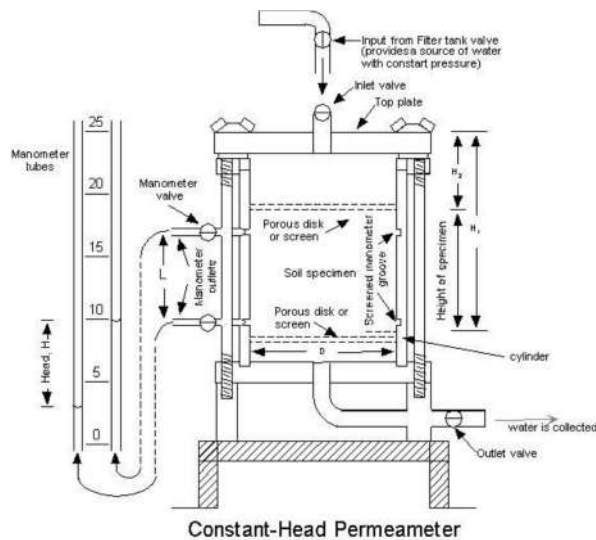
Theory –

The property of the soils which permits water (fluids) to percolate through its continuously connected voids is called its permeability. Depending upon the value of Reynolds number the flow of water through soils may be 'laminar' or 'turbulent'. In laminar flow, a particle of water starting from a given position follows a definite path without crisscrossing the path of other particles. In turbulent flow the particles do not follow any definite path but have random, twisting and crisscrossing path. For laminar and steady flow, according to Darcy's law the rate of flow of water is proportional to the hydraulic gradient in uniform and homogeneous soils.

e. $V \propto I$, where v = discharge velocity of water

i = hydraulic gradient

Procedure-



- 1) Remove the cover of the mould and apply a little grease on the sides of the mould.
- 2) Weigh the mould with dummy plate.
- 3) Measure the internal diameter and effective height of the mould, then attach the collar and the base plate.
- 4) (i) Compact the soil at given dry density and moisture content by a suitable static or dynamic device for remolded samples.
(ii) For undisturbed samples, trim off the undisturbed specimen in the form of a cylinder about 85 mm in diameter and height equal to that of mould. Place the specimen centrally over the bottom porous disc & filter paper. Fill the annular space between the mould and the specimen with an impervious material such as cement slurry or bentonite slurry to provide sealing against leakage from the sides.
5. Remove the collar & base plate, trim off the excess soil and level with the top of the mould.
6. Clean the outside of the mould and dummy plate.
7. Disconnect the reservoirs from the outlet as the bottom and connect to the inlet at the top plate.
8. Open stop cock at the cover and allow water to flow out so that all the air in the cylinder is removed.
9. When all the air has escaped close the stop cock and open the outlet. Allow the water to flow through the soil and established a steady flow.
10. When steady flow is reached collect the water in a measuring flask for a convenient time interval. Repeat this thrice quantity of water collected must be same, otherwise observations are repeated.
11. Repeat step (5) for atleast two more different time intervals.

Observations and Calculations-

$$kT = 2.303 [aL/At] \log_{10} (h_1/h_2)$$

For Constant Head Method

$$KT = Q.L/At.h$$

Where, KT = Coefficient of Permeability at Test temperature T0C (cm/sec) Q = Quantity of Water collected in time t (ml)

L = length of soil sample (cm)

A= Cross-Sectional area of Soil Sample (cm²) h = Constant hydraulic head (cm)

Report the coefficient of permeability at standard temperature of 270c

$$k_{27} = kT \times [\eta T / \eta_{27}]$$

Where, k₂₇= Coefficient of Permeability at 270c kT = Coefficient of permeability at T0c

ηT = Coefficient of Viscosity at T0c

η_{27} = Coefficient of Viscosity at 270c

Calculate the density, void ratio, and degree of saturation at which test is performed Void ratio, e = $[G_s/\gamma_d] - 1$

Where, G_s = Specific Gravity of soil grains (Assume 2.6 if not known) γ_d = Dry density of soil sample (gm/cc)

Degree of Saturation, S = $(w \times G_s) / e$

Where, w = Moisture content (%)

Table for constant head method-

Determination No.	At the start (before saturation)			At the end(after saturation)		
Container No.						
Mass of Container + Wet soil (gm)						
Mass of Container + dry soil (gm)						
Mass of Container (gm)						
Mass of soil (3-4) (gm)						
Mass of water (2-3) (gm)						
Water Content, w = $[6/5] \times 100$						
Degree of Saturation = $(w \times G_s) / e$						

Length of sample, L (cm)=

Diameter of sample, d(cm)=

Area of sample, A(cm²)=

Mass of Mould + dummy plate, W₁ (gm) =

Mass of Mould + soil + dummy plate, W₂ (gm) =

Mass of soil, W= (W₁-W₂) (gm) =

Volume of Soil sample (cm³) =

Density of soil sample $[\gamma_t = W/V]$ (gm/cc) =

Moisture Content at the start, w₁ =

Dry Density of soil sample $[\gamma_d = (\gamma_t / (1 + w_1))]$ (gm/cc) =

Void ratio, e= $[G_s/\gamma_d] - 1$ =

Determination No.	1	2	3
Hydraulic Head, h (cm)			

Time Interval (sec)			
Quantity of flow, Q (ml)			
I test for time , t (ml)			
II test for same time , t (ml)			
III test for same time , t (ml)			
Average Quantity of Flow (A+B+C)/3 (ml)			
Coefficient of permeability at test temperature k_T (cm/sec)			
Coefficient of permeability at 27deg temperature k_{27} (cm/sec)			

Result-

Average value of coefficient of permeability at test temperature, $K_T =$

Average value of coefficient of permeability at standard temperature 27c , $K_{27} =$

Void ratio of soil sample, $e =$

Type of soil =

EXPERIMENT NO :- 8

Aim – To perform Permeability test using Falling-head test method.

Apparatus:-

1. Permeameter mould (internal dia = 100 m.m. effective height = 127.3 m.m. Capacity 100cc)
2. Accessories of the permeameter (cover, base, detachable collar, porous stones, dummy plate) (common)
3. Round filter paper 100 mm. dia (common)
4. A static or dynamic compaction device (if remolded samples are used)
5. Constant head reservoir (common)
6. Graduated glass stand pipe (internal dia 5 to 20 mm, preferably 10mm) (Variable head)
7. Support frame and clamps (variable head)
8. Funnel (variable head)
9. Measuring flask (Constant head)
10. Metre Scale
11. Balance
12. Stop watch
13. Thermometer
14. Deaired water
15. Drying oven
16. IS: Sieve 4.75 mm
17. Grease
18. Straight edge
19. Drying crucibles
20. Desiccator

Theory –

The property of the soils which permits water (fluids) to percolate through its continuously connected voids is called its permeability. Depending upon the value of Reynolds number the flow of water through soils may be 'laminar' or 'turbulent'. In laminar flow, a particle of water starting from a given position follows a definite path without crisscrossing the path of other particles. In turbulent flow the particles do not follow any definite path but have random, twisting and crisscrossing path. For laminar and steady flow, according to Darcy's law the rate of flow of water is proportional to the hydraulic gradient in uniform and homogeneous soils.

Procedure-

1. Remove the cover of the mould and apply a little grease on the sides of the mould.
2. Weigh the mould with dummy plate.
3. Measure the internal diameter and effective height of the mould, then attach the collar and the base plate.
4. (i) Compact the soil at given dry density and moisture content by a suitable static or dynamic device for remolded samples.
(ii) For undisturbed samples, trim off the undisturbed specimen in the form of a cylinder about 85 mm in diameter and height equal to that of mould. Place the specimen centrally over the bottom porous disc & filter paper. Fill the annular space between the mould and the specimen with an impervious material such as cement slurry or bentonite slurry to provide sealing against leakage from the sides.
5. Remove the collar & base plate, trim off the excess soil and level with the top of the mould.
6. Clean the outside of the mould and dummy plate.
7. Weigh the mould with soil and dummy plate. Difference of this mass and the mass taken in step 2 will give the mass of soil used.
8. Apply grease around the porous stone and base plate, put the porous stone inside the base plate and filter paper on porous stone.
9. Remove the dummy plate and place the mould with washer on the base plate. 10. Put the small quantity of the soil sample in drying oven to determine the moisture content.
11. Clean the edges of the mould and the collar and apply grease in the grooves around them.
12. Place a filter paper, porous stone and washer on the top of the soil sample and fix up the collar again.
13. Connect the reservoir with water to the outlet at the bottom of the mould and allow the water to flow in. wait till the water has been able to travel up and saturate the sample. Allow about one cm depth of free water to collect on the top of the sample.
14. Fill the remaining portion of the cylinder with deaired water without disturbing the surface of the soil.
15. Fix the cover plate over the collar and tighten the nuts in the rods.
16. Disconnect the reservoir from the outlet at the bottom and connect the stand pipe to the inlet at the top plate. Fill the stand pipe with water.
17. Open the stop cock at the top and allow water to flow out so that all the air in the cylinder is removed.
18. Fix the height h_1 and h_2 on the pipe from the centre of the outlet such that $(h_1 - h_2)$ is about 30 to 40 cm. Mark the level of $\sqrt{h_1 h_2}$ from the centre of the outlet.
19. When all the air has escaped, close the stop cock and allow the water from the pipe to flow through the soil and establish a speedy flow.
20. Record the time intervals for the head to fall from $\sqrt{h_1 h_2}$ to h_2 . The time intervals should be same, otherwise steady flow is not established.

21. Change the height h_1 and h_2 and record the time intervals. 22. Stop the flow of water, disconnect all parts.

23. Take a small quantity of the soil sample from the mould in the drying crucible and put inside the drying oven for moisture content determination.

24. Measure the temperature of the water.

Observation and calculation-

Diameter of stand pipe (cm) =

Cross sectional area of pipe (cm²) =

Temperature of water, T_0 =

Correction factor due to temperature, $C_t = \eta T / \eta_{27}$ =

Constant Factor = $2.303 [aL/A]$ =

Determination No.	1	2	3
Initial Head, h_1 (cm)			
Final Head, h_2 (cm)			
Head $\sqrt{h_1 - h_2}$ (cm)			
Time Interval (sec)			
From h_1 to $\sqrt{h_1 - h_2}$			
From $\sqrt{h_1 - h_2}$ to h_2			
From h_1 to h_2 , $t = A + B$			
Log ₁₀ h_1/h_2			
Coefficient of permeability k (cm/sec) K at test temperature T_0 = $(2.303 [aL/A] \times 5) / 4C$			
Coefficient of permeability k (cm/sec) K at test temperature 27°C = $6 \times \eta T / \eta_{27}$			

Result-

Average value of coefficient of permeability at test temperature, K_T =

Average value of coefficient of permeability at standard temperature 27°C , K_{27} =

Void ratio of soil sample, e =

Type of soil =

EXPERIMENT NO :- 9

Aim – To Determine optimum moisture content and maximum dry density of the soil by Standard Proctor test method.

Apparatus:-

1. Cylindrical mould (capacity 1000 cc, internal dia 100 mm. effective height 127.3 mm)
Cylindrical mould (capacity 2250cc, internal dia 150 mm. effective height 127.30m.m)
2. Rammer for light compaction (face diameter 50 mm mass of 2.6 kg free drop 310 mm) or
Rammer for heavy compaction (face diameter 50 mm mass of 4.89 kg free drop 450 mm)
3. Mould accessories (detachable base plate removable collar)
4. Balance
5. Drying oven
6. Desiccators
7. Graduated jar
8. Straight edge
9. Large mixing pan
10. Spatula
11. Scoop

Procedure-

- 1) Take about 20 kg for 100cc mould or 45 kg for 2250 cc mould of air dried soil
- 2) Sieve this through 20mm and 4.75 mm sieves
- 3) Calculate the percentage retained on 20 mm and 4.75mm sieves and passing from 4.75 mm sieve.
- 4) Mix the soil retained on 4.75 mm sieve and passing from 4.75 mm sieve thoroughly in the proportion obtained in step3
- 5) Take about 2.5kg of the soil for 1000 cc mould or take about 2.8 kg or the soil for 1000 cc mould or 6.5kg for 2250 cc mould for heavy compaction.
- 6) Add water to it bring its moisture content to about 4% in coarse grained soils and 8% in fine grained soils
- 7) Clean, dry and grease lightly the mould and base plate. Weigh the mould with base plate.
- 8) Fit the collar and place the mould on a solid base.
- 9) For light compaction, compact wet soil in three equal layers by the rammer of mass 2.6 kg and free fall 31 cm with 25 evenly distributed blows in each layer for 10 cm diameter mould and 56 blows for 15cm diameter mould. Alternatively for heavy compaction compact the soils using the rammer of mass 4.89 kg and free fall 45 cm in five layers. Each layer being given 25 blows for 10 cm diameter mould and 56 blows for 15 cm diameter mould.
- 10) Remove the collar and trim off the soil flush with the top of the mould. In removing the collar rotate it to break the bond between it and the soil before lifting it off the mould.
- 11) Clean the outside of the mould and base plate, weigh the mould with soil and base plate.

- 12) Remove the soil from the mould and obtain a representative soil sample from the bottom middle and top for water content determination.
- 13) Weigh the drying crucible with samples and put samples and put in the drying oven at temperature 105°C to 110°C or 24 hours.
- 14) Repeat the above procedure with 7,10,13,16,19,22% of water contents on coarse grained fresh soil samples and 11,14,17,20,23 and 26% of water contents of fine grained fresh soil samples approximately.

Precautions

1. Adequate period is allowed for mixing the water with soil before compaction
2. The blows should be uniformly distributed over the surface of each layer.
3. Each layer of compacted soil is scored with a spatula before placing the soil for the succeeding layer.
4. The amount of soil used be just sufficient to fill the mould i.e. at the last layer the Surface of the soil should be slightly (5mm) above the top rim of the mould.
5. Mould should be placed on a solid foundation during compaction.

Observation & calculation-

Soil retained on 20mm sieve (%) =
 Soil retained on 4.75mm sieve (%) =
 Soil passing from 4.75mm sieve (%) =

Determination No.	1	2	3
Mass of mould + compacted soil (gm)			
Mass of compacted soil Wt (gm)			
Wet Density, $\gamma_t = W_t / V$ (gm/cc)			
Crucible No.			
Mass of Crucible + Wet soil (gm)			
Mass of Crucible + dry soil (gm)			
Mass of water (5-6)			
Mass of Crucible (gm)			
Mass of dry soil (gm) (6-8)			
Water Content, w (%) = $\frac{W}{D} \times 100$			
Dry Density, $\gamma_d = \gamma_t / (1+w)$ (gm/cc)			
Dry Density at 100% Saturation (gm/cc)			

Result-

Optimum Moisture Content =----- %
 Maximum Dry Density = ----- gm/cc
 Degree of Saturation at OMC=----- %

EXPERIMENT NO :- 10

Aim – To determine the consolidation properties of a given soil.

Apparatus:-

1. Consolidometer
2. Loading device consisting of a frame, lever system, loading yoke, steel ball, dial gauge fixing device, and weights.
3. Dial gauge (accuracy of 0.01 mm)
4. thermostatically controlled oven
5. stopwatch
6. sample extractor,
7. balance,
8. soil. trimming tools
9. spatula
10. filter papers
11. sample containers
12. wash bottles.

NEED AND SCOPE:

This test simulates one-dimensional primary consolidation with double drainage. The following parameters are determined by conducting a Consolidation test on fine-grained soils: a. Pressure-void ratio relationship b. Compression and Recompression index c. Coefficient of consolidation at various pressures d. Pre consolidation pressure e. Degree of consolidation at any time f. Rate of consolidation under vertical loads The above information can be used to predict the time rate and extent of settlement of structures founded on fine-grained soils. It is also helpful in analyzing the stress history of soil.

SAMPLE PREPARATION:

1. Undisturbed sample: From the sample tube (Shelby tube), eject the sample into the consolidation ring. The sample should project about one cm from the outer ring. Trim the sample smoothly and flush with the top and bottom of the ring by using a wire saw. Clean the ring from the outside and keep it ready for weighing.

2. Remolded sample:

- a. Choose the density and water content at which the sample has to be compacted from the moisture-density curve, and calculate the quantity of soil and water required to mix and compact.
- b. Compact the specimen in compaction mould in three layers using the standard rammers (moist tamping technique).
- c. Eject the specimen from the mould using the sample extractor

PROCEDURE:

1. Saturate two porous stones either by boiling them in distilled water for about 15 minutes or by keeping them submerged in the distilled water for 4 to 8 hrs. Fittings of the Consolidometer, which is to be enclosed, shall be moistened.
2. Assemble the Consolidometer with the soil specimen and porous stones at the top and bottom of the specimen, and provide a filter paper between the soil specimen and porous stone.
3. Position the pressure pad centrally on the top porous stone. Mount the mould assembly on the loading frame, and center it such that the load applied is axial. Make sure that the porous stone and pressure pad are not touching the walls of mould on their sides.
4. Position the dial gauge to measure the vertical compression of the specimen. The dial gauge holder should be set so that the dial gauge is at the beginning of its release run, and also allowing sufficient margin for the swelling of the soil, if any.
5. Fill the mould with water and apply an initial load to the assembly. The magnitude of this load should be chosen by trial, such that there is no swelling. It should be not less than 0.05 kg/cm^2 for ordinary soils & 0.025 kg/cm^2 for very soft soils. The load should be allowed to stand until there is no change in dial gauge readings for two consecutive hours or for a maximum of 24 hours.
6. Note the final dial reading under the initial load. Apply the first load of intensity 0.1 kg/cm^2 (Approx.) and start the stopwatch simultaneously. Record the dial gauge readings at various time intervals. The dial gauge readings are taken until 90% consolidation is reached. Primary consolidation is gradually reached within 24 hrs.
7. At the end of the period specified above, take the dial reading and time reading. Double the load intensity and take the dial readings at various time intervals. Repeat this procedure for successive load increments. The usual loading intensity is as follows (approx.): 0.1, 0.2, 0.5, 1, 2, 4 and 8 kg/cm^2 . Dial gauge reading with time should be recorded for each loading increment.

8. On completion of the final loading stage, the specimen shall be unloaded by pressure decrements, which decrease the load to one-fourth of the last load. Dial gauge readings may be taken as necessary during each stage of unloading. If desired, the time intervals used during the consolidation increments may be adopted; usually, it is possible to proceed much more rapidly (IS 2720- Part 15).

9. In the unloading phase, the load needs to be reduced in the reverse order and allowed to stand for at least 2 hrs or until the dial gauge reading becomes constant. Take the final reading of the dial gauge.

10. Quickly dismantle the specimen assembly and remove the excess water on the soil specimen in the oven; note its dry weight.

CALCULATIONS:

1. Height of solids (HS) is calculated from the equation $HS = WS / (GS \cdot \gamma_w \cdot A)$

2. Void ratio. The void ratio at the end of various pressures is calculated from the equation $e = (H - HS) / HS$

3. Coefficient of consolidation. The Coefficient of consolidation at each pressure increment is calculated by using the following equations:

i. $C_v = 0.197 d^2 / t_{50}$ (Log fitting method)

ii. $C_v = 0.848 d^2 / t_{90}$ (Square fitting method)

4. Compression Index. To determine the compression index, a plot of voids ratio (e) Vs log (t) is made. The virgin compression curve would be a straight line and the slope of this line would give the compression index C_c .

5. Coefficient of compressibility. It is calculated as follows-

6. Coefficient of permeability. It is calculated as follows-

GRAPHS:

1. Dial reading Vs log of time or
2. Dial reading Vs square root of time
3. Voids ratio (e) Vs effective vertical stress

OBSERVATION AND READING (LOADING):

Table I: Data Sheet for Consolidation Test:

Time Displacement Relationship Ring Dimensions: Diameter (cm): _____ Area (cm²): _____
 Height (cm): -----
 Initial Data: Specimen Ht (cm). _____
 Specific Gravity of Soil: _____
 Weight of wet soil + Ring (g): _____
 Weight of Ring (g): _____
 Bulk Density (g/cc): _____

OBSERVATION TABLE-1

Pressure intensity (kg/cm ²)	0.1	0.2	0.5	1	2	4	8
Time (min)							
0							
0.25							
1							
2							
4							
8							
15							
30							
1hr							
2hr							
4hr							
8hr							
24hr							

OBSERVATION AND READING (UNLOADING):

Removed Pressure (kg/cm ²)	Retained Pressure (kg/cm ²)	Dial gauge reading
0	8	
4	4	
2	2	
1	1	
0.5	0.5	
0.3	0.2	
0.1	0.1	
0.1	0.05 (Seating pressure)	

Water Content Determination:

Weight of Saturated Sample + Ring (g): _____

Weight of oven-dried soil +Ring (g): _____

Water Content (%): _____

Table II: Data Sheet for Consolidation Test: Pressure-Voids Ratio

[illegible]



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B-TECH - 4TH SEMESTER

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